The Canoe Traveller's Network

The instructor's key / Ordinary tandem team / Double carries / Model assumptions, not observed travel times

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I. THREE PRE-DESIGNED JOURNEYS

A - Inner circuit / 13.25-27.25 h

Clearwater - West Pike - Pine - Little Caribou - Caribou - Clearwater

B - Eastern circuit / 19.5-40.75 h

Clearwater - West Pike - East Pike - John - Little John - McFarland
Pine - Little Caribou - Caribou - Clearwater

C - Rose out and back / 17.25-33.25 h

Clearwater - Mountain - Watap - Rove - Rose; return by the same chain

Totals are moving + handling hours, not days or daily targets.
Rest, camps, queues, difficult weather and windbound waits add time.
Both directions use the same broad scenario envelope; wind direction
changes which end of the range is plausible.

II. HOW THE HOURS ARE RECKONED

Paddle: geographic water-path distance, plus 15-30% indirect travel;
20-35% on more winding paths. Divide by 4 km/h (favourable) or
2 km/h (adverse but travelable). These are teaching assumptions.

Carry: 2 loaded walks at 2-3 km/h, 1 return at 3-4 km/h,
plus 10-20 minutes of handling per connection. One rod = 5.0292 m.

Bounds round outward to quarter hours. A 0-0.25 h label means
a brief nonzero leg. All rod counts are derived, not official counts.

III. READING THE SPARSE CHART

Each lake has at most one pair of time pies:
the longest modeled landing-to-landing crossing.
Left = shorter time; right = longer time.
A full black disc = one fixed 15-hour period.
Its exact endpoint pair is recorded in the layer
and the table below. Shorter crossings may take
less time. A lake without pies has no modeled
crossing. All alternatives remain in the register.

Bent carries echo recorded trail geometry. Small
IDs identify them; rods are kept in the CSV.
No inference should be made from an absent link.

Landmark leaders associate a feature with a lake,
not a measured sight line. Water-visible features
have published lake-view evidence. (walk) marks
waterfalls reached on foot, not a canoe sighting.

Landmark coordinates and viewing descriptions
are in the geographic project and LANDMARKS.md.
Generic DEM summit candidates are hidden there;
they are not shown as visible landmarks here.

IV. THE HYPOTHETICAL EXERCISE

S marks the Clearwater departure. The three
original trips remain in the key. Extra lakes widen
the study area; they do not certify a through-route.
All time ranges are scenarios, not predictions.
The diagram cannot supply bearings or distances.

SOURCES & REPRODUCTION

MN DNR waters / USFS trails / published lake views
Friends of BWCA digitized loop / Parks Canada history
Sources: docs/SOURCES.md and docs/LANDMARKS.md
Assumptions and original geometry are bundled.

The Canoe Traveller's Network

The reckoning of each passage / Source-backed distances; modeled hours / All distances below are geographic

ID	Lake / connection	Points	Mode	Base km	Extra paddle	Hours	Evidence
a	Clearwater / West Pike		carry	1.04	-	1-1.75 h	approximate*
b	West Pike / Pine		carry	1.62	-	1.5-2.5 h	USFS + shore
c	West Pike / East Pike		carry	0.92	-	1-1.75 h	USFS + shore
d	East Pike / John		carry	1.07	-	1-2 h	USFS + shore
e	John / Little John		carry	0.08	-	0-0.5 h	USFS + shore
f	Little John / McFarland		river	0.03	-	0-0.25 h	approximate*
g	McFarland / Pine		carry	0.05	-	0-0.5 h	USFS + shore
h	Pine / Little Caribou		carry	0.40	-	0.5-1 h	approximate*
i	Little Caribou / Caribou		carry	0.10	-	0.25-0.5 h	approximate*
j	Caribou / Clearwater		carry	1.05	-	1-1.75 h	USFS + shore
k	Clearwater / Mountain		carry	0.42	-	0.5-1 h	USFS + shore
l	Mountain / Watap		carry	0.47	-	0.5-1 h	USFS + shore
n	Rove / Rose		carry	3.28	-	3-4.75 h	USFS + shore
o	Caribou / Deer		carry	0.24	-	0.25-0.75 h	USFS + shore
p	Pine / Canoe		carry	2.32	-	2.25-3.5 h	USFS + shore
q	Canoe / Alder		carry	0.13	-	0.25-0.75 h	USFS + shore
r	Canoe / Crystal		carry	0.26	-	0.25-0.75 h	USFS + shore
s	Rose / Duncan		carry	0.39	-	0.5-1 h	USFS + shore
t	Duncan / Bearskin		carry	0.43	-	0.5-1 h	USFS + shore
u	Mountain / Fan		carry	0.36	-	0.5-1 h	USFS + shore
v	Fan / Vaseux		carry	0.23	-	0.25-0.75 h	USFS + shore
w	Vaseux / Moose		carry	0.75	-	0.75-1.5 h	USFS + shore
m	Watap / Rove		narrows	<0.01	-	0-0.25 h	approximate*
A01	Little North / Little Gunflint		carry	0.09	-	0-0.5 h	USFS + shore
A02	Alder / East Bearskin		carry	0.26	-	0.25-0.75 h	USFS + shore
A03	Aspen / East Bearskin		carry	0.51	-	0.5-1.25 h	USFS + shore

** Approximate passage sketches or short water links. Full source records, exact hours and assumptions: data/leg-times.csv.*

Paddle points use connecting carry IDs. S = departure; T = imagined turn. Only the longest crossing is drawn as pies per lake.

The Canoe Traveller's Network

The reckoning of each passage / Source-backed distances; modeled hours / All distances below are geographic

ID	Lake / connection	Points	Mode	Base km	Extra paddle	Hours	Evidence
A04	Hungry Jack / Flour		carry	0.71	-	0.75-1.5 h	USFS + shore
A05	Skipper / Poplar		carry	1.62	-	1.5-2.5 h	USFS + shore
A06	Little Trout / Rum		carry	0.32	-	0.25-1 h	USFS + shore
A07	Flour / Aspen		carry	0.54	-	0.5-1.25 h	USFS + shore
A08	South / Rat		carry	0.40	-	0.5-1 h	USFS + shore
A09	Moss / Duncan		carry	0.28	-	0.25-0.75 h	USFS + shore
A10	Partridge / Duncan		carry	0.84	-	0.75-1.5 h	USFS + shore
A11	Bearskin / Hungry Jack		carry	0.11	-	0.25-0.5 h	USFS + shore
A12	Henson / Gaskin		carry	0.46	-	0.5-1 h	USFS + shore
A13	Swallow / Pillsbery		carry	0.45	-	0.5-1 h	USFS + shore
A14	Caribou (Poplar) / Horseshoe		carry	0.12	-	0.25-0.5 h	USFS + shore
A15	Lower George / Karl		carry	0.17	-	0.25-0.75 h	USFS + shore
A16	Rum / Kroft		carry	0.27	-	0.25-0.75 h	USFS + shore
A17	Lux / Carl		carry	0.07	-	0-0.5 h	USFS + shore
A18	Rush / One Island		carry	0.33	-	0.25-1 h	USFS + shore
A19	Rush / Little Rush		carry	0.22	-	0.25-0.75 h	USFS + shore
A20	Horseshoe / Vista		carry	0.14	-	0.25-0.75 h	USFS + shore
A21	Meeds / Swallow		carry	0.55	-	0.5-1.25 h	USFS + shore
A22	Flour / East Bearskin		carry	0.43	-	0.5-1 h	USFS + shore
A23	Flour / Moon		carry	0.48	-	0.5-1 h	USFS + shore
A24	Little Rush / Skipper		carry	0.09	-	0.25-0.5 h	USFS + shore
A25	Omega / Henson		carry	0.18	-	0.25-0.75 h	USFS + shore
A26	Horseshoe / Allen		carry	0.28	-	0.25-0.75 h	USFS + shore
A27	North / South		carry	0.42	-	0.5-1 h	USFS + shore
A28	Kroft / Ram		carry	0.35	-	0.25-1 h	USFS + shore
A29	Pillsbery / Henson		carry	0.27	-	0.25-0.75 h	USFS + shore

** Approximate passage sketches or short water links. Full source records, exact hours and assumptions: data/leg-times.csv.*

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The Canoe Traveller's Network

The reckoning of each passage / Source-backed distances; modeled hours / All distances below are geographic

ID	Lake / connection	Points	Mode	Base km	Extra paddle	Hours	Evidence
A30	Omega / Winchell		carry	0.25	-	0.25-0.75 h	USFS + shore
A31	Portage / One Island		carry	0.83	-	0.75-1.5 h	USFS + shore
A32	Rose / Rat		carry	0.04	-	0-0.5 h	USFS + shore
A33	Poplar / Meeds		carry	1.46	-	1.5-2.5 h	USFS + shore
A34	Moon / East Bearskin		carry	0.52	-	0.5-1.25 h	USFS + shore
A35	East Bearskin / Crocodile		carry	0.62	-	0.5-1.25 h	USFS + shore
A36	Missing Link / Snipe		carry	0.67	-	0.75-1.25 h	USFS + shore
A37	Muskeg / Kiskadinna		carry	0.90	-	0.75-1.75 h	USFS + shore
A38	Gaskin / Winchell		carry	0.29	-	0.25-0.75 h	USFS + shore
A39	Round / Missing Link		carry	0.71	-	0.75-1.5 h	USFS + shore
A40	Allen / Jump		carry	0.10	-	0.25-0.5 h	USFS + shore
A41	Cross Bay / Rib		carry	0.30	-	0.25-0.75 h	USFS + shore
A42	Horseshoe / Gaskin		carry	0.50	-	0.5-1.25 h	USFS + shore
A43	Karl / Long Island		carry	0.20	-	0.25-0.75 h	USFS + shore
A44	Winchell / Wanihigan		carry	0.07	-	0-0.5 h	USFS + shore
A45	Kiskadinna / Omega		carry	0.19	-	0.25-0.75 h	USFS + shore
A46	Tuscarora / Missing Link		carry	1.82	-	1.75-3 h	USFS + shore
A47	Jump / Gaskin		carry	0.10	-	0.25-0.5 h	USFS + shore
A48	Poplar / Lizz		carry	0.25	-	0.25-0.75 h	USFS + shore
A49	Iron / Portage		carry	0.16	-	0.25-0.75 h	USFS + shore
A50	Allen / Pillsbery		carry	0.47	-	0.5-1 h	USFS + shore
A51	Misquah / Little Trout		carry	1.02	-	1-1.75 h	USFS + shore
W01	Clearwater	S-a	paddle	9.82	15-30%	2.75-6.5 h	water model
W02	Clearwater	S-j	paddle	1.97	15-30%	0.5-1.5 h	water model
W03	Clearwater	S-k	paddle	7.54	15-30%	2-5 h	water model
W04	Clearwater	a-j	paddle	8.48	20-35%	2.5-5.75 h	water model

** Approximate passage sketches or short water links. Full source records, exact hours and assumptions: data/leg-times.csv.*

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ID	Lake / connection	Points	Mode	Base km	Extra paddle	Hours	Evidence
W05	Clearwater	a-k	paddle	2.54	15-30%	0.5-1.75 h	water model
W06	Clearwater	j-k	paddle	6.23	15-30%	1.75-4.25 h	water model
W07	West Pike	a-b	paddle	6.07	15-30%	1.5-4 h	water model
W08	West Pike	a-c	paddle	6.07	15-30%	1.5-4 h	water model
W09	West Pike	b-c	paddle	0.00	15-30%	same landing	water model
W10	East Pike	c-d	paddle	5.83	15-30%	1.5-4 h	water model
W11	John	d-e	paddle	2.09	15-30%	0.5-1.5 h	water model
W12	Little John	e-f	paddle	0.90	15-30%	0.25-0.75 h	water model
W13	McFarland	f-g	paddle	4.00	20-35%	1-2.75 h	water model
W14	Pine	b-g	paddle	5.49	15-30%	1.5-3.75 h	water model
W15	Pine	b-h	paddle	6.12	15-30%	1.75-4 h	water model
W16	Pine	b-p	paddle	6.21	15-30%	1.75-4.25 h	water model
W17	Pine	g-h	paddle	11.51	15-30%	3.25-7.5 h	water model
W18	Pine	g-p	paddle	11.61	15-30%	3.25-7.75 h	water model
W19	Pine	h-p	paddle	0.13	15-30%	0-0.25 h	water model
W20	Little Caribou	h-i	paddle	1.00	15-30%	0.25-0.75 h	water model
W21	Caribou	i-j	paddle	5.52	15-30%	1.5-3.75 h	water model
W22	Caribou	i-o	paddle	5.37	15-30%	1.5-3.5 h	water model
W23	Caribou	j-o	paddle	0.22	15-30%	0-0.25 h	water model
W24	Mountain	k-l	paddle	1.02	20-35%	0.25-0.75 h	water model
W25	Mountain	k-u	paddle	10.97	15-30%	3-7.25 h	water model
W26	Mountain	l-u	paddle	11.38	15-30%	3.25-7.5 h	water model
W27	Watap	l-m	paddle	3.26	15-30%	0.75-2.25 h	water model
W28	Rove	m-n	paddle	1.91	15-30%	0.5-1.25 h	water model
W29	Rose	A32-T	paddle	6.10	15-30%	1.75-4 h	water model
W30	Rose	A32-n	paddle	7.66	15-30%	2-5 h	water model

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W31	Rose	A32-s	paddle	5.03	20-35%	1.5-3.5 h	water model
W32	Rose	T-n	paddle	1.57	15-30%	0.25-1.25 h	water model
W33	Rose	T-s	paddle	2.19	15-30%	0.5-1.5 h	water model
W34	Rose	n-s	paddle	3.71	15-30%	1-2.5 h	water model
W35	Deer	T-o	paddle	0.33	15-30%	0-0.25 h	water model
W36	Canoe	p-q	paddle	1.33	15-30%	0.25-1 h	water model
W37	Canoe	p-r	paddle	0.71	20-35%	0-0.5 h	water model
W38	Canoe	q-r	paddle	0.74	15-30%	0-0.5 h	water model
W39	Crystal	T-r	paddle	1.82	15-30%	0.5-1.25 h	water model
W40	Duncan	A09-A10	paddle	1.99	20-35%	0.5-1.5 h	water model
W41	Duncan	A09-s	paddle	2.71	15-30%	0.75-2 h	water model
W42	Duncan	A09-t	paddle	0.95	15-30%	0.25-0.75 h	water model
W43	Duncan	A10-s	paddle	3.66	15-30%	1-2.5 h	water model
W44	Duncan	A10-t	paddle	2.40	15-30%	0.5-1.75 h	water model
W45	Duncan	s-t	paddle	2.09	15-30%	0.5-1.5 h	water model
W46	Bearskin	A11-t	paddle	1.15	15-30%	0.25-0.75 h	water model
W47	Fan	u-v	paddle	0.33	15-30%	0-0.25 h	water model
W48	Vaseux	v-w	paddle	0.37	15-30%	0-0.25 h	water model
W49	Moose	T-w	paddle	3.40	15-30%	0.75-2.25 h	water model
W50	Hungry Jack	A04-A11	paddle	2.20	15-30%	0.5-1.5 h	water model
W51	Moss	A09-T	paddle	1.00	20-35%	0.25-0.75 h	water model
W52	Partridge	A10-T	paddle	1.09	15-30%	0.25-0.75 h	water model
W53	Rat	A08-A32	paddle	0.52	15-30%	0-0.5 h	water model
W54	South	A08-A27	paddle	4.26	15-30%	1-3 h	water model
W55	North	A27-T	paddle	1.65	15-30%	0.25-1.25 h	water model
W56	Little Gunflint	A01-T	paddle	1.19	15-30%	0.25-1 h	water model

** Approximate passage sketches or short water links. Full source records, exact hours and assumptions: data/leg-times.csv.*

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The reckoning of each passage / Source-backed distances; modeled hours / All distances below are geographic

ID	Lake / connection	Points	Mode	Base km	Extra paddle	Hours	Evidence
W57	Little North	A01-T	paddle	0.93	15-30%	0.25-0.75 h	water model
W58	Iron	A49-T	paddle	3.06	15-30%	0.75-2 h	water model
W59	Portage	A31-A49	paddle	1.05	20-35%	0.25-0.75 h	water model
W60	One Island	A18-A31	paddle	0.70	15-30%	0-0.5 h	water model
W61	Rush	A18-A19	paddle	0.18	20-35%	0-0.25 h	water model
W62	Little Rush	A19-A24	paddle	0.87	15-30%	0-0.75 h	water model
W63	Poplar	A05-A33	paddle	2.61	20-35%	0.75-2 h	water model
W64	Poplar	A05-A48	paddle	2.47	15-30%	0.5-1.75 h	water model
W65	Poplar	A33-A48	paddle	1.86	20-35%	0.5-1.5 h	water model
W66	Caribou (Poplar)	A14-T	paddle	1.23	15-30%	0.25-1 h	water model
W67	Horseshoe	A14-A20	paddle	2.78	20-35%	0.75-2 h	water model
W68	Horseshoe	A14-A26	paddle	1.36	15-30%	0.25-1 h	water model
W69	Horseshoe	A14-A42	paddle	2.88	20-35%	0.75-2 h	water model
W70	Horseshoe	A20-A26	paddle	3.92	20-35%	1-2.75 h	water model
W71	Horseshoe	A20-A42	paddle	3.44	20-35%	1-2.5 h	water model
W72	Horseshoe	A26-A42	paddle	4.04	20-35%	1-2.75 h	water model
W73	Lizz	A48-T	paddle	0.56	15-30%	0-0.5 h	water model
W74	Meeds	A21-A33	paddle	3.18	15-30%	0.75-2.25 h	water model
W75	Swallow	A13-A21	paddle	0.09	15-30%	0-0.25 h	water model
W76	Pillsbery	A13-A29	paddle	0.42	15-30%	0-0.5 h	water model
W77	Pillsbery	A13-A50	paddle	1.30	15-30%	0.25-1 h	water model
W78	Pillsbery	A29-A50	paddle	1.72	15-30%	0.25-1.25 h	water model
W79	Allen	A26-A40	paddle	0.74	15-30%	0-0.5 h	water model
W80	Jump	A40-A47	paddle	0.40	15-30%	0-0.5 h	water model
W81	Gaskin	A12-A38	paddle	2.89	20-35%	0.75-2 h	water model
W82	Gaskin	A12-A42	paddle	4.55	15-30%	1.25-3 h	water model

** Approximate passage sketches or short water links. Full source records, exact hours and assumptions: data/leg-times.csv.*

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ID	Lake / connection	Points	Mode	Base km	Extra paddle	Hours	Evidence
W83	Gaskin	A12-A47	paddle	3.84	15-30%	1-2.5 h	water model
W84	Gaskin	A38-A42	paddle	2.46	20-35%	0.5-1.75 h	water model
W85	Gaskin	A38-A47	paddle	1.75	20-35%	0.5-1.25 h	water model
W86	Gaskin	A42-A47	paddle	0.95	20-35%	0.25-0.75 h	water model
W87	Henson	A12-A25	paddle	3.76	15-30%	1-2.5 h	water model
W88	Henson	A12-A29	paddle	0.39	15-30%	0-0.5 h	water model
W89	Henson	A25-A29	paddle	3.50	15-30%	1-2.5 h	water model
W90	Omega	A25-A30	paddle	0.76	15-30%	0-0.5 h	water model
W91	Omega	A25-A45	paddle	2.54	15-30%	0.5-1.75 h	water model
W92	Omega	A30-A45	paddle	2.30	20-35%	0.5-1.75 h	water model
W93	Winchell	A30-A38	paddle	7.13	15-30%	2-4.75 h	water model
W94	Winchell	A30-A44	paddle	2.04	20-35%	0.5-1.5 h	water model
W95	Winchell	A38-A44	paddle	7.00	15-30%	2-4.75 h	water model
W96	Wanihigan	A44-T	paddle	0.56	15-30%	0-0.5 h	water model
W97	Kiskadinna	A37-A45	paddle	3.28	15-30%	0.75-2.25 h	water model
W98	Muskeg	A37-T	paddle	0.52	15-30%	0-0.5 h	water model
W99	Long Island	A43-T	paddle	3.57	20-35%	1-2.5 h	water model
W100	Karl	A15-A43	paddle	0.27	15-30%	0-0.25 h	water model
W101	Cross Bay	A41-T	paddle	1.44	15-30%	0.25-1 h	water model
W102	Rib	A41-T	paddle	0.92	20-35%	0.25-0.75 h	water model
W103	Lower George	A15-T	paddle	0.47	15-30%	0-0.5 h	water model
W104	Missing Link	A36-A39	paddle	0.67	20-35%	0-0.5 h	water model
W105	Missing Link	A36-A46	paddle	0.78	15-30%	0-0.75 h	water model
W106	Missing Link	A39-A46	paddle	0.72	15-30%	0-0.5 h	water model
W107	Tuscarora	A46-T	paddle	1.24	15-30%	0.25-1 h	water model
W108	Round	A39-T	paddle	0.63	15-30%	0-0.5 h	water model

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ID	Lake / connection	Points	Mode	Base km	Extra paddle	Hours	Evidence
W109	East Bearskin	A02-A03	paddle	5.71	15-30%	1.5-3.75 h	water model
W110	East Bearskin	A02-A22	paddle	2.38	15-30%	0.5-1.75 h	water model
W111	East Bearskin	A02-A34	paddle	3.56	20-35%	1-2.5 h	water model
W112	East Bearskin	A02-A35	paddle	2.81	15-30%	0.75-2 h	water model
W113	East Bearskin	A03-A22	paddle	3.77	15-30%	1-2.5 h	water model
W114	East Bearskin	A22-A34	paddle	1.60	15-30%	0.25-1.25 h	water model
W115	East Bearskin	A34-A35	paddle	2.46	15-30%	0.5-1.75 h	water model
W116	Flour	A04-A07	paddle	0.89	15-30%	0.25-0.75 h	water model
W117	Flour	A04-A22	paddle	4.08	15-30%	1-2.75 h	water model
W118	Flour	A04-A23	paddle	4.15	15-30%	1-2.75 h	water model
W119	Flour	A07-A22	paddle	3.50	20-35%	1-2.5 h	water model
W120	Flour	A07-A23	paddle	3.58	15-30%	1-2.5 h	water model
W121	Flour	A22-A23	paddle	1.82	20-35%	0.5-1.25 h	water model
W122	Aspen	A03-A07	paddle	0.61	20-35%	0-0.5 h	water model
W123	Moon	A23-A34	paddle	1.02	15-30%	0.25-0.75 h	water model
W124	Crocodile	A35-T	paddle	2.29	15-30%	0.5-1.5 h	water model
W125	Vista	A20-T	paddle	0.64	15-30%	0-0.5 h	water model
W126	Lux	A17-T	paddle	0.54	15-30%	0-0.5 h	water model
W127	Carl	A17-T	paddle	0.55	15-30%	0-0.5 h	water model
W128	Misquah	A51-T	paddle	0.40	15-30%	0-0.5 h	water model
W129	Little Trout	A06-A51	paddle	0.70	15-30%	0-0.5 h	water model
W130	Rum	A06-A16	paddle	0.76	15-30%	0-0.5 h	water model
W131	Kroft	A16-A28	paddle	0.51	15-30%	0-0.5 h	water model
W132	Ram	A28-T	paddle	0.31	15-30%	0-0.25 h	water model

** Approximate passage sketches or short water links. Full source records, exact hours and assumptions: data/leg-times.csv.*

Paddle points use connecting carry IDs. S = departure; T = imagined turn. Only the longest crossing is drawn as pies per lake.